IMEN574 / IDSC574 · PROGRAMMING FOR AX

20

Vision-Language Models

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Where We Are

01 Programming with AI

Lectures 2–4

02 Python and Classical Machine Learning

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Vision-Language Models

01 **Aligning Modalities**

Build a shared space

02 **Querying Visual Content**

Use text as a task

03 **Generating Grounded Answers**

Connect pixels to language

01 · Aligning Modalities

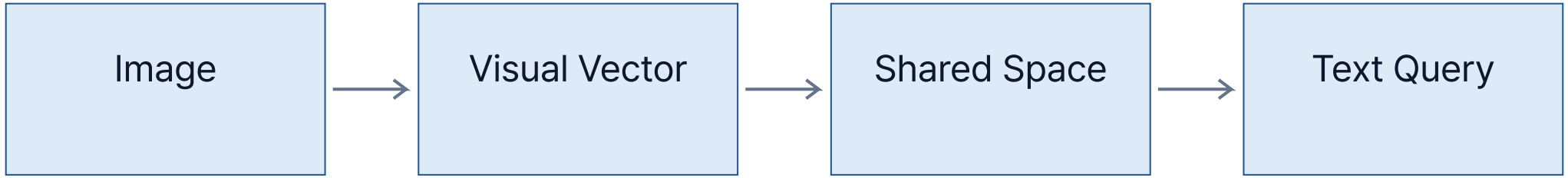
01 **Aligning Modalities**
Build a shared space

02 **Querying Visual Content**
Use text as a task

03 **Generating Grounded Answers**
Connect pixels to language

Why Connect Modalities

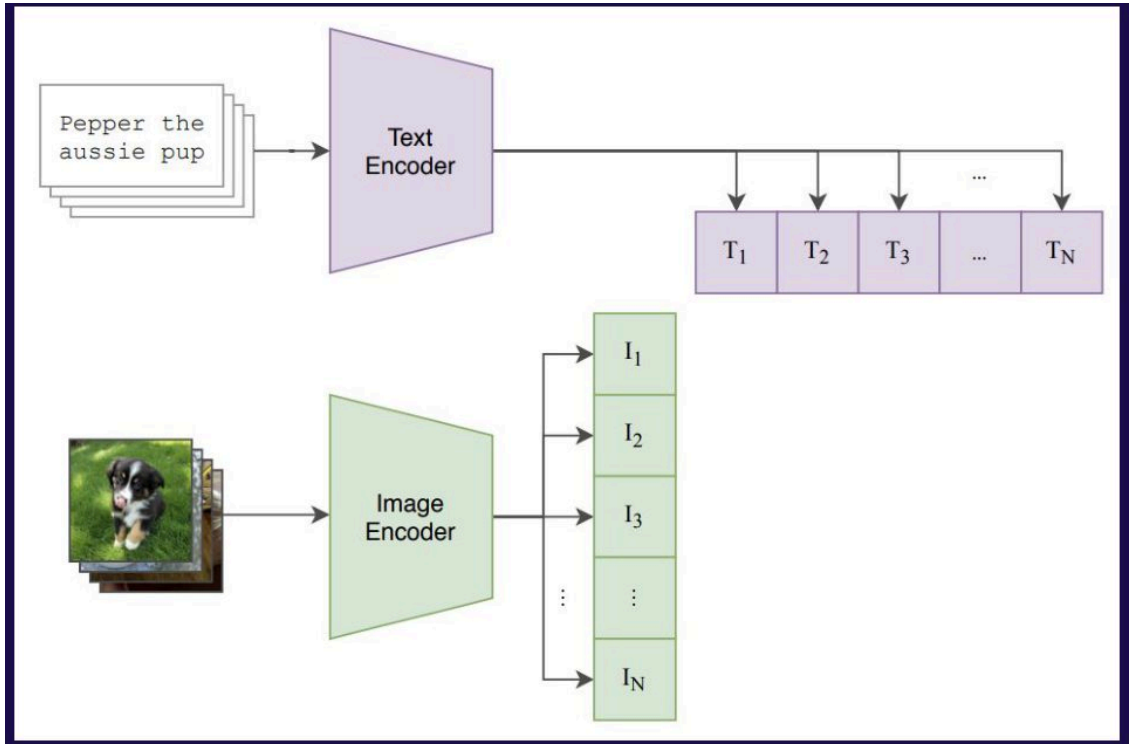
An image vector needs a semantic coordinate system



A text query needs a bridge into visual features

Paired Image and Text

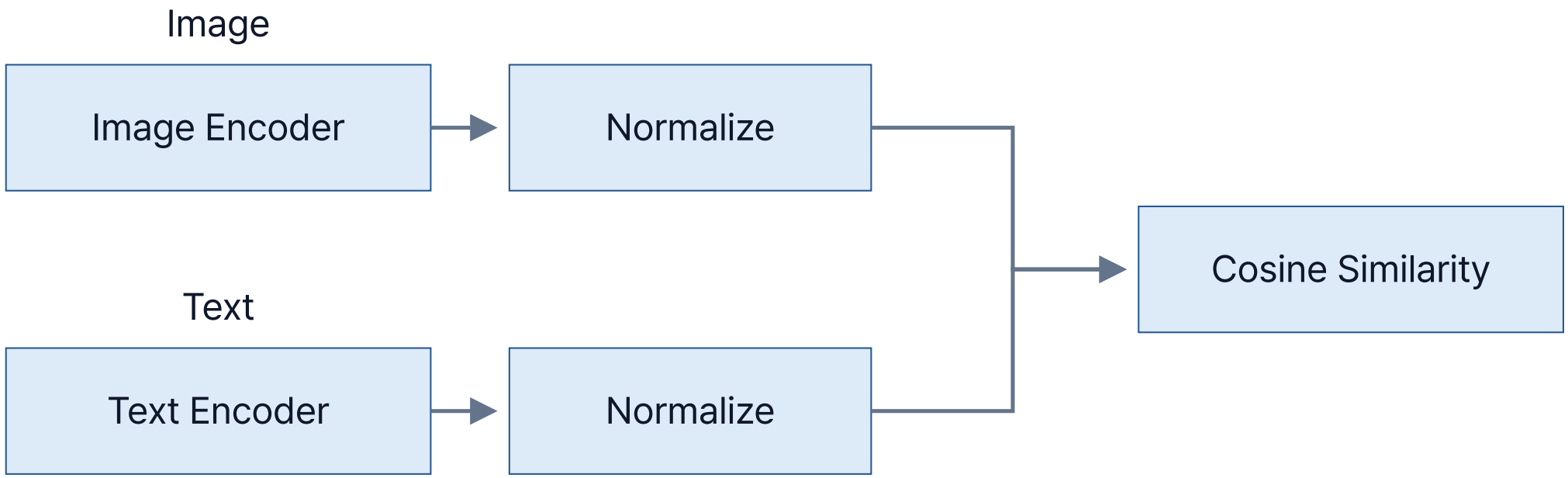
Captions provide a weak semantic relation



Paired observations connect two input modalities

Dual Encoders

Each modality maps into one comparable vector space



Separate encoders can share a comparison space

https://github.com/openai/CLIP

Cross-Modal Similarity

A matrix compares every image with every text

	Text A	Text B	Text C
Image A	0.9	0.2	0.1
Image B	0.1	0.8	0.3
Image C	0.2	0.1	0.95

Matched image-text pairs lie on the diagonal

Symmetric Contrastive Loss

Both retrieval directions contribute training signal

$$L = (L_{image \to text} + L_{text \to image}) / 2$$

A Small Retrieval Example

Normalized vectors make the ranking explicit

Image query	1	0
Text: dog	0.8	0.6
Text: car	0	1

The dog vector ranks above the car vector

Worked Example: Hard-Negative Retrieval

Trace the inputs before execution

```
Toy query = [1,.2,0]; normalize query and gallery
dog=[1,0,0], cat=[0,1,0], wolf=[.98,.2,0], car=[0,0,1]
Rank unit-vector dot products; top k=2.
```

Demo: Hard-Negative Retrieval

Executable educational calculation; no pretrained inference

Remove the wolf candidate; compare the ranking and explain what changed

Query dog axis 1 Query cat axis 0.2 Candidate wolf Included Top k 2

Toy text query → fixed image feature gallery



Top-2: wolf, dog.
Questions: Does removing a hard negative change the task? Is rank a calibrated probability?

Answer: Hard-Negative Retrieval

Same inputs as the initial demo state

```
Top2 wolf .999992, dog .980581
cat .196116, car 0
```

Failure: Hard-Negative Retrieval

Identify the assumption that breaks

```
Candidate removal makes the task easier without improving an encoder
cosine is not a calibrated probability
```

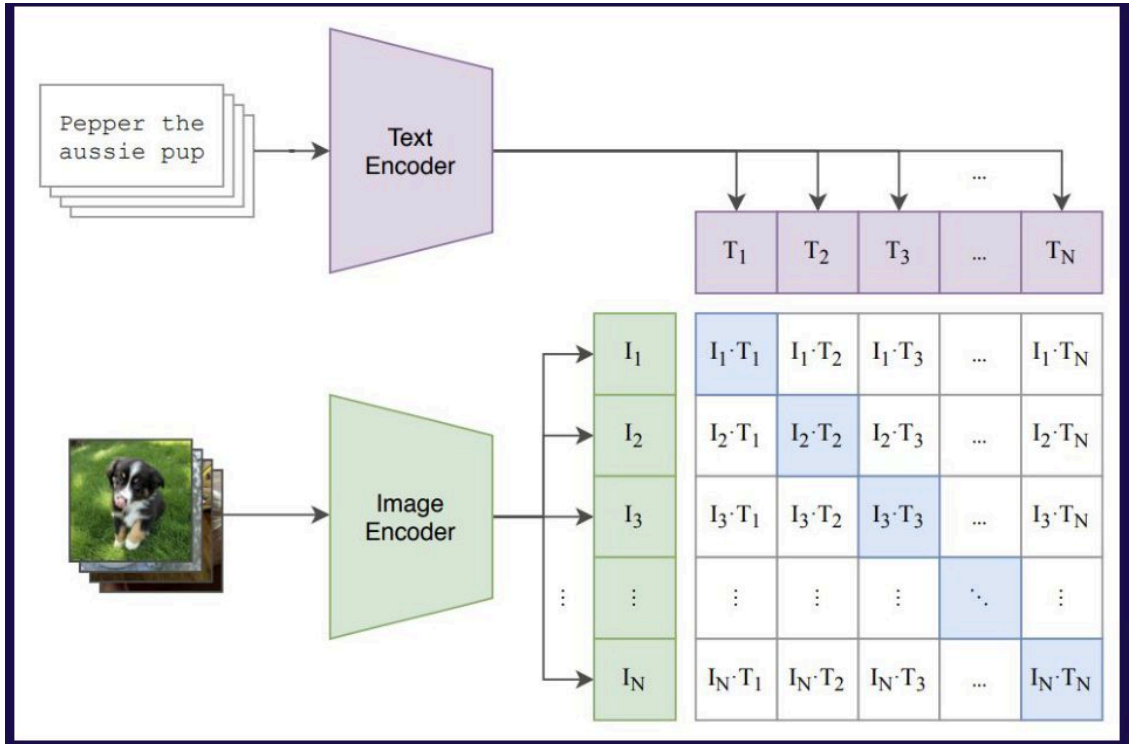
Negative Pair Ambiguity

A mismatched caption may still describe the image

Image	Caption	Pair label
Two dogs	A dog outdoors	Matched
Two dogs	Animals on grass	Unmatched

CLIP Training Matrix

The original design learns from matched pairs at batch scale



Both modalities participate in the same contrastive matrix

https://arxiv.org/abs/2103.00020

Caveat: A Shared Space

Similarity measures compatibility rather than factual truth

Item	Field 1	Field 2
Ranking	Image-text similarity	high
Verification	Exact count verified	no

Related descriptions can disagree on precise details

02 · Querying Visual Content

01 Aligning Modalities

Build a shared space

02 **Querying Visual Content**

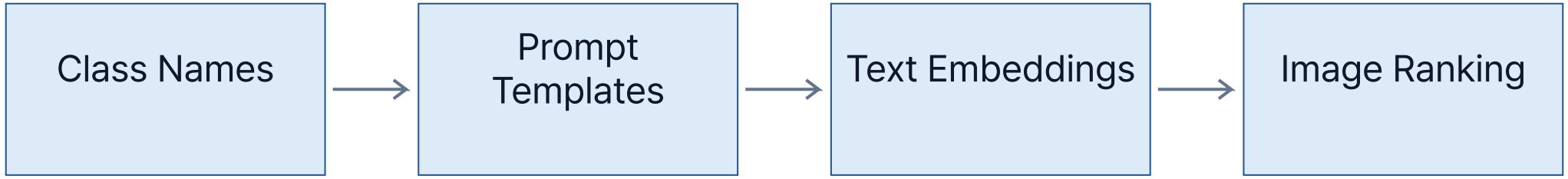
Use text as a task

03 Generating Grounded Answers

Connect pixels to language

Why Zero-Shot Classification

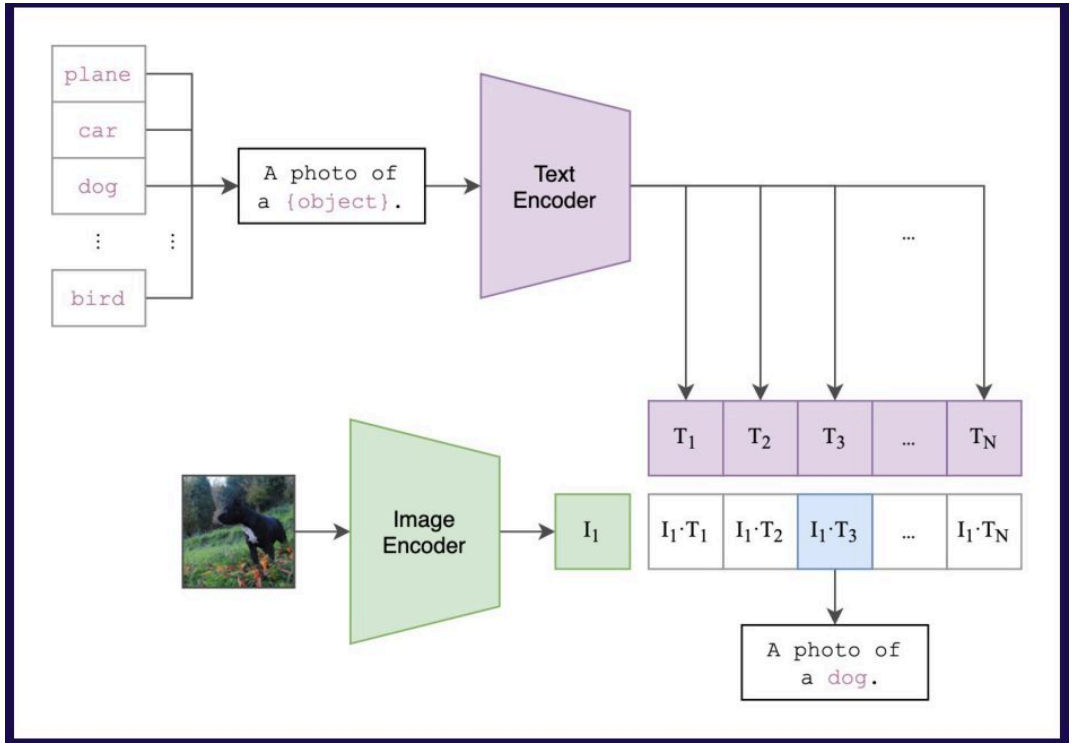
New labels can be expressed as text queries



Text labels replace a newly trained classifier head

Text-Based Classifier

Candidate descriptions form the class prototypes



The label list defines the available decisions

https://arxiv.org/abs/2103.00020

Prompt Templates

Wording changes the vector used for a class

Class	Template
dog	a photo of a dog
dog	a sketch of a dog
dog	a microscopic image of a dog

Template Averaging

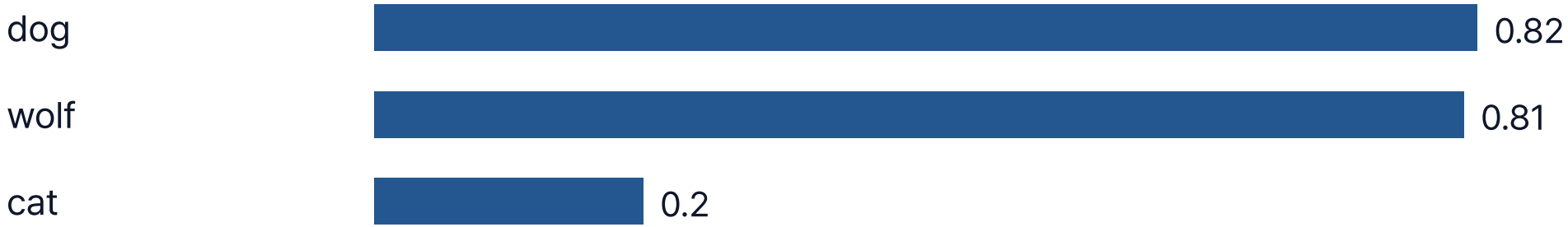
Multiple descriptions can form a class prototype

Template A	0.9	0.1
Template B	0.8	0.3
Mean	0.85	0.2

A prototype can aggregate several descriptions

Candidate-Set Sensitivity

Removing a label can force a different answer



Illustrative image-text similarity

The highest score depends on which labels are offered

Demo: CLIP Prompt and Candidate Sets

Actual dual-encoder similarities on Oxford Pets

Code | Results

Run in Colab ↗

```
images = [Image.open(image_paths[0]).convert("RGB")]
candidates = ["a photo of a dog", "a photo of a cat", "a photo of a bird"]
probabilities, predictions = clip_zero_shot_classify(images, candidates)
print(dict(zip(candidates, probabilities[0].round(3))))
print("Top candidate:", candidates[predictions[0]])
```

Change the recorded prompt set. Does a high score remain high after adding a candidate?

Softmax scores depend on the candidate set

Unknown-Class Rejection

A rejection rule must be validated for the deployment setting



Rejection adds a decision beyond top-one ranking

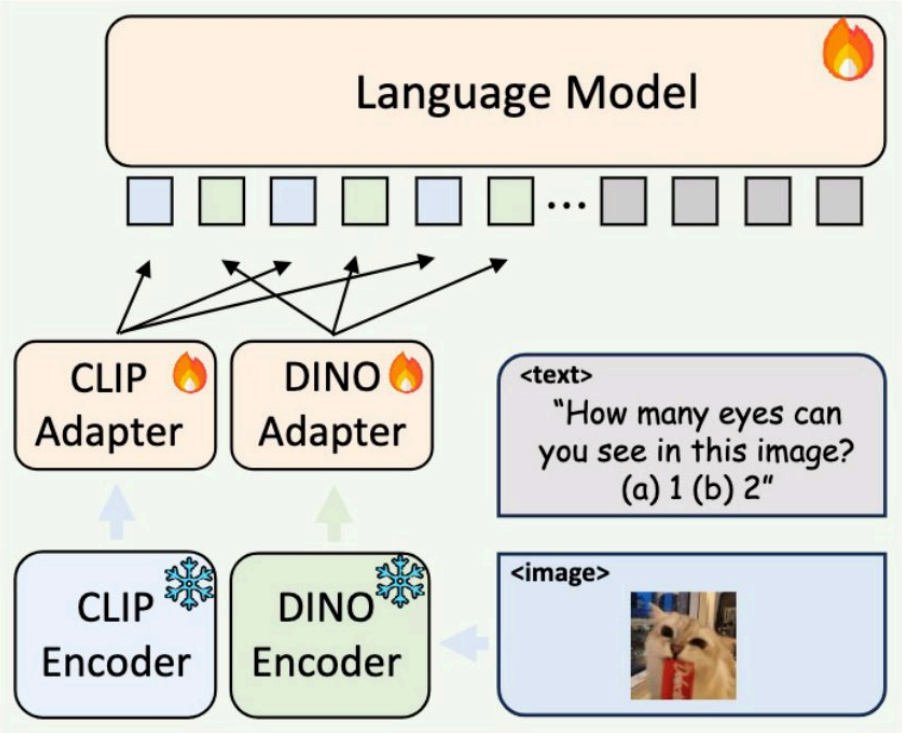
Visual and Semantic Cues

Image-only features and language alignment capture different relations

Cue	Training relation
Visual	Two views of one image
Semantic	An image and its caption

Complementary Representations

Separate visual and semantic encoders can feed a shared task



Different training signals can supply complementary features

Demo: CLIP Geometry and Retrieval

Measured CLIP retrieval ranks

Code | Results

Run in Colab ↗

```
query = "a fluffy white cat"
query_vector = get_clip_text_embeddings([query])
scores = (query_vector @ all_img_embeds.T).ravel()
ranking = np.argsort(scores)[::-1][:5]
print([(image_paths[i], float(scores[i])) for i in ranking])
```

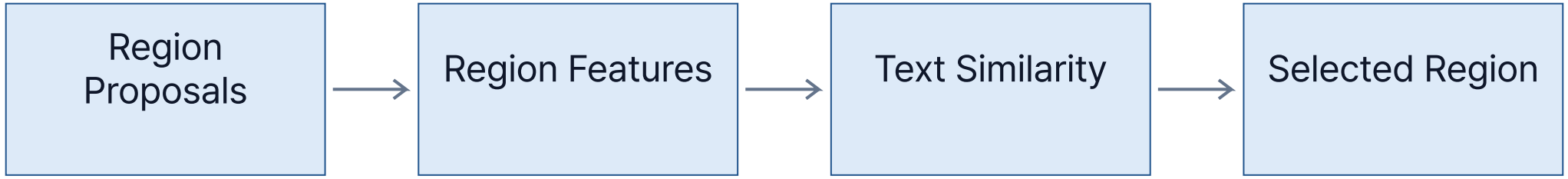
Compare queries and returned images. Which visual cue changes the rank?

A retrieval rank does not establish the presence of every requested detail

https://huggingface.co/openai/clip-vit-base-patch32

Region-Text Retrieval

Pooling the whole image can dilute a small target



Region features preserve the query target

Worked Example: Region Grounding

Trace the inputs before execution

```
Toy region features: A=[1,.2], B=[1,-.2], C=[.1,1]
animal query=[1,0], left=[0,1], left animal=[1,1]
L2-normalize, rank regions, return top1 xyxy box.
```

Demo: Region Grounding

Executable educational calculation; no pretrained inference

Compare animal with left animal; inspect which features make the regions different

Text query animal Region top k 1

Fixed region features: A=[1,.2], B=[1,−.2], C=[.1,1]

Region A: 0.981

xyxy [20,30,120,150]

Returned 1 region(s), ranked by cosine. Ties retain region order.
Questions: Can “animal” separate A from B? Does “left” encode geometry without a spatial feature?

Answer: Region Grounding

Same inputs as the initial demo state

```
animal ties A and B at .980581
stable order returns A [20,30,120,150]
```

Failure: Region Grounding

Identify the assumption that breaks

```
A feature name does not confer spatial understanding
fixed toy features do not run CLIP, DINO or a grounding model
```

Grounding and Alignment

A text-conditioned detector answers a spatial query

Query	Output
Which image matches?	Image index
Which object matches?	Bounding box
Which pixels match?	Mask

Retrieval Evaluation

Use rank metrics with a fixed gallery and relevance rule

	Rank 1	Rank 2	Rank 3
Relevant 1	1	0	0
Relevant 2	0	1	0
Relevant 3	0	0	1

Retrieval quality depends on the relevance definition

03 · Generating Grounded Answers

01 Aligning Modalities

Build a shared space

02 Querying Visual Content

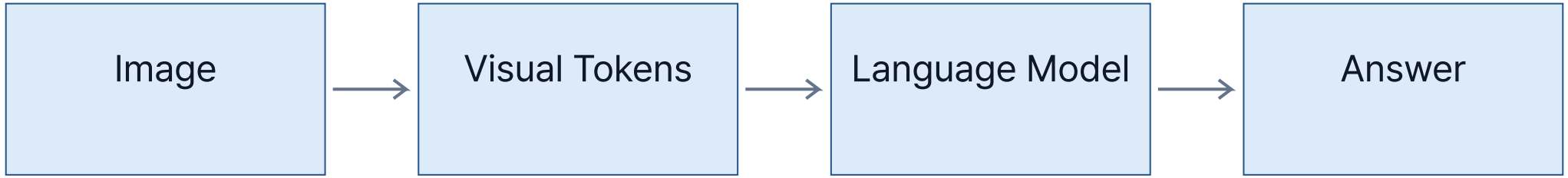
Use text as a task

03 **Generating Grounded Answers**

Connect pixels to language

Why Generate from Images

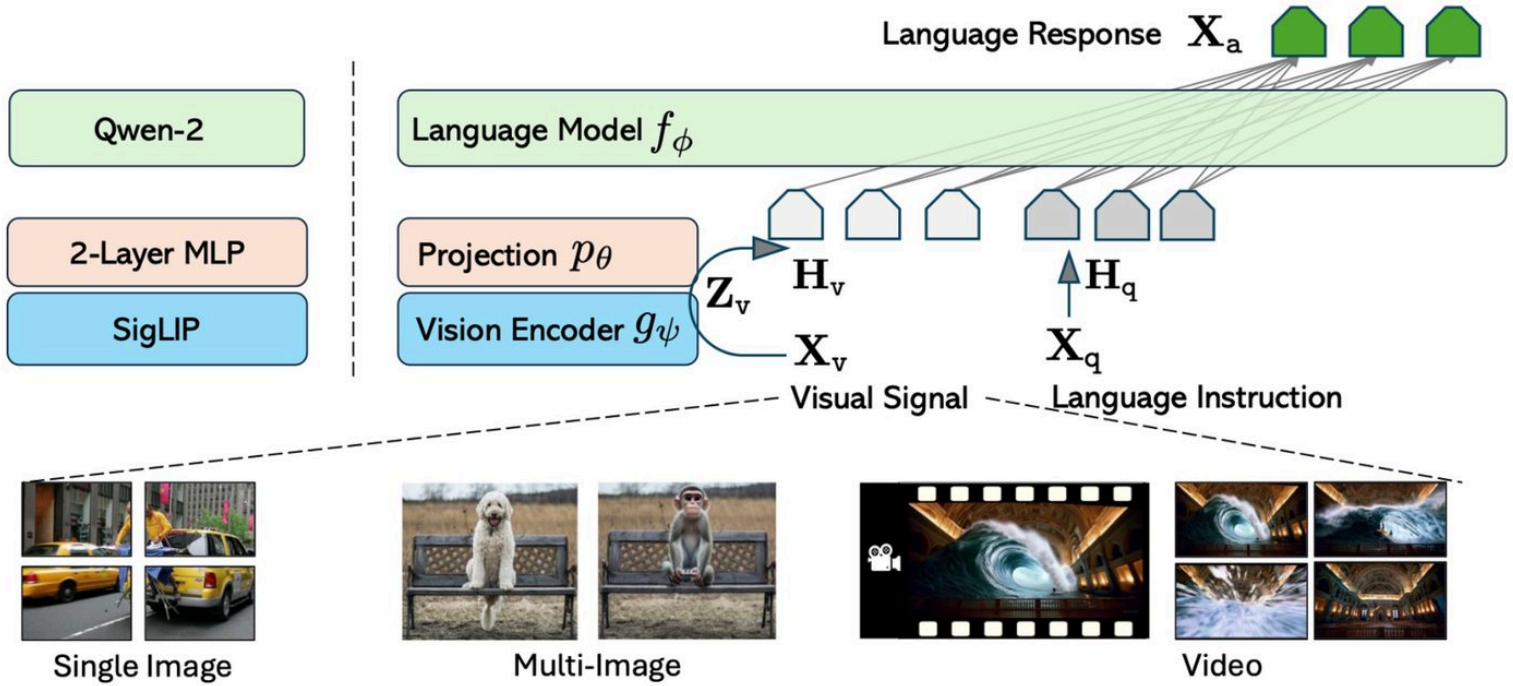
Some questions require more than choosing a fixed label



Visual tokens can condition an open-ended answer

Visual-Language Connector

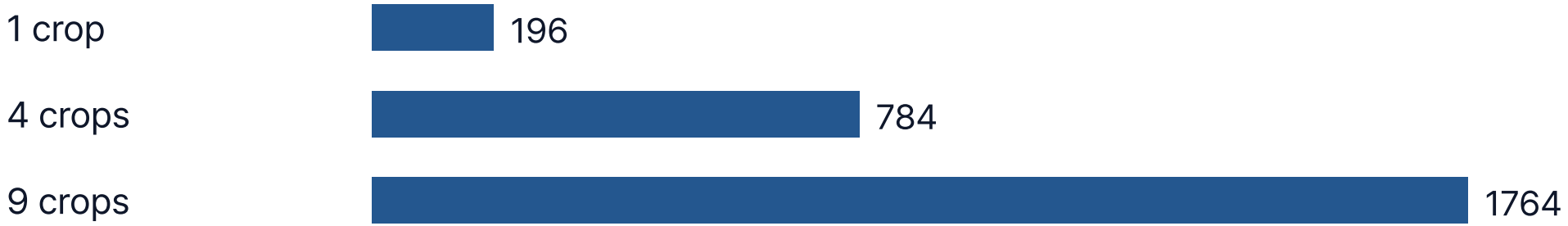
A connector links visual features to a language model



A connector places image features into the language input

Visual Token Budget

Image resolution and cropping change the context footprint



Illustrative tokens at 196 per crop

More crops increase the visual context size

Instruction Data

Questions and answers teach how to use visual evidence

Input	Target
Image + describe	A scene description
Image + locate	A spatial answer
Image + compare	An attribute relation

OCR and Fine Detail

A resized view may lose the evidence before reasoning begins

Item	Field 1	Field 2
Input	Original text	8 pixels high
Resize	Downscaled text	2 pixels high

Missing pixels cannot be recovered by a better prompt

Counting and Relations

Correct object names do not imply correct quantities

Claim	Required evidence
Three valves	Every distinct instance
Left of the pump	A defined viewpoint
Damaged handle	Visible defect detail

Caveat: Visual Hallucination

A plausible scene description may add absent objects

Item	Field 1	Field 2
Evidence	Observed	pump, pipe
Answer	Generated	pump, pipe, gauge

Generated nouns must be checked against the image

Demo: Qwen3-VL Answers

Qwen3-VL · Image-conditioned inference

Code | Results

Run in Colab ↗

```
question = "What animal is visible? Describe the background."
answer = ask_vlm(image_paths[0], question, max_new_tokens=48)
print(answer)
```

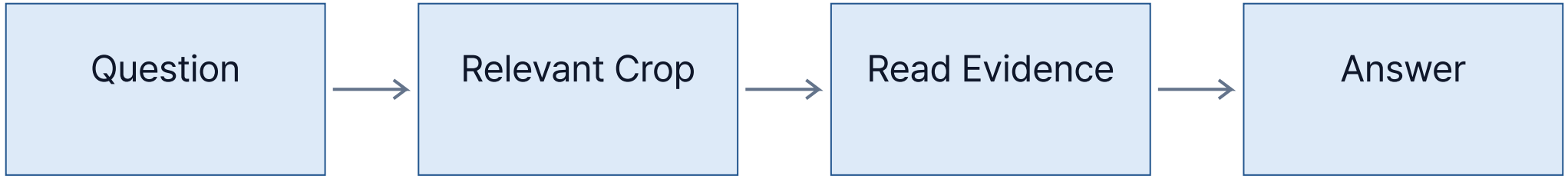
Choose an experiment. Which claims are directly supported by the image?

Recorded model responses are evidence to inspect, not verified facts

https://huggingface.co/Qwen/Qwen3-VL-2B-Instruct

Evidence-Grounded Interaction

Crop, inspect and answer only the supported question



Targeted inspection reduces unsupported detail

Frozen Embedding Code

Normalization and gallery alignment are explicit operations

```
images = images / np.linalg.norm(images, axis=1, keepdims=True)
texts = texts / np.linalg.norm(texts, axis=1, keepdims=True)
similarity = images @ texts.T
ranked = np.argsort(-similarity, axis=1)
```

Multimodal Evaluation

Separate retrieval, perception and answer correctness

Layer	Measure
Retrieval	Relevant item at rank k
Perception	Object or region correctness
Answer	Supported claims only

Takeaway: Vision-Language Models

01 Shared embeddings support ranking across modalities

02 Candidate labels and prompts change the decision rule

03 Open-ended answers need image-grounded verification